

Lab 1 - Setting Up the IoT Security Laboratory Environment

Intended Learning Outcome

Set up and verify a functional IoT security laboratory environment while demonstrating safe and ethical laboratory practices.

Engagement Update

The client has authorized the Assessment Cell to prepare the laboratory environment that will be used throughout the security assessment engagement. Before security assessments can begin, all hardware, software, and supporting services must be installed, assembled, configured, and validated to establish a reliable and repeatable testing environment.

The laboratory environment will be prepared over three consecutive sessions, with each session building upon the previous one.

Assessment Objectives

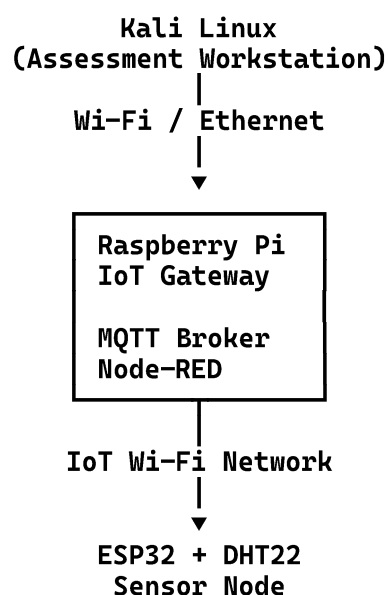
Upon completion of this laboratory exercise, students should be able to:

- Install and configure the Kali Linux assessment workstation.
- Assemble and program the ESP32 IoT device.
- Deploy and configure the Raspberry Pi gateway.
- Establish communication between IoT devices and supporting services.
- Verify end-to-end connectivity within the IoT security testbed.
- Document the baseline laboratory configuration.

Laboratory Overview

Week	Laboratory Activity	Expected Output
Week 1	Prepare the Assessment Workstation	Kali Linux Workstation
Week 2	Assemble and Configure the IoT Device	Operational ESP32 Sensor Node
Week 3	Deploy the Gateway and Validate the Testbed	Fully Functional IoT Security Laboratory

Conceptual Laboratory Topology



Part A – Prepare the Kali Linux Assessment Workstation

Overview

The Kali Linux assessment workstation will serve as the primary platform for performing reconnaissance, vulnerability assessment, network analysis, and evidence collection throughout this course. In this activity, you will install Oracle VirtualBox, deploy the provided Kali Linux virtual machine, and verify that the workstation is operational before proceeding to subsequent laboratory exercises.

Expected Outcome

At the end of this activity, you should have:

- Oracle VirtualBox successfully installed.
- Kali Linux virtual machine deployed.
- Internet connectivity verified.
- A fully operational assessment workstation ready for IoT security testing.

Required Resources

Hardware	Software
Student Laptop	Kali Linux
Raspberry Pi	Raspberry Pi Imager
ESP32 Development Board	Arduino IDE
DHT22 Sensor	MQTT Explorer
USB Cables	Balena Etcher / Pi Imager
Wi-Fi Network	Web Browser

Step 1. Install Oracle VirtualBox

Oracle VirtualBox is the virtualization software used to host the Kali Linux virtual machine.

1. Open a web browser and navigate to the official VirtualBox website.
<https://www.virtualbox.org>
2. Select **Downloads**.
3. Download the installation package appropriate for your operating system.
4. After the download is complete, execute the installer.
5. Complete the installation using the default installation options unless otherwise instructed by your instructor.

Important

Download VirtualBox only from its official website to ensure that the software has not been modified or tampered with.

Step 2. Obtain the Kali Linux Virtual Machine

To ensure that all Assessment Cells use a consistent environment, a pre-configured Kali Linux virtual machine can be downloaded at <https://cdimage.kali.org/kali-2026.2/kali-linux-2026.2-virtualbox-amd64.7z>

1. Access the link using a web browser.
2. Download the Kali Linux **OVA** appliance.
3. Save the downloaded file to a known location on your computer.

Record the location of the downloaded file.

Item	Observation
OVA Filename	
Download Location	

Step 3. Add the Kali Linux Virtual Machine

1. Launch **Oracle VirtualBox**.
2. Locate the downloaded Kali Linux VirtualBox archive (.7z) on your computer.
3. Extract the downloaded archive using an archive utility such as **7-Zip**.
4. Open Oracle VirtualBox and select:
Machine → Add
5. Browse to the folder where you extracted the Kali Linux image.
6. Select the **Kali Linux .vbox file** and click **Open**.
7. The Kali Linux virtual machine should now appear in the VirtualBox Manager.
8. Select the Kali Linux virtual machine and review its configuration.
9. If your computer has sufficient system memory, your instructor may recommend increasing the allocated RAM.
10. **Do not change the network settings** unless instructed by your instructor. The laboratory activities may require a specific network configuration.
11. Proceed to **Step 4: Start the Kali Linux Virtual Machine**.

Important: Do not select **File → Import Appliance**. The pre-built Kali Linux VirtualBox image is added directly through **Machine → Add** after extracting the downloaded archive.

Step 4. Start the Kali Linux Virtual Machine

1. Select the imported Kali Linux virtual machine.
2. Click **Start**.
3. Wait until the Kali Linux desktop environment loads completely.
4. Log in using the credentials provided by your instructor.

Step 5. Verify the Assessment Workstation

Validation Item	Status
Kali Linux boots successfully	☐
Desktop environment loads correctly	☐
Terminal application operational	☐
Internet connectivity verified	☐
DNS resolution verified	☐

Open a terminal and verify network connectivity.

```
ping -c 4 8.8.8.8
```

Four successful replies should be displayed with 0% packet loss.

Next, verify that DNS resolution is functioning correctly.

```
ping -c 4 google.com
```

The domain name should resolve successfully and return four replies.

If both commands complete successfully, the assessment workstation is ready for use.

Lab 1 - Part A Completion Record

Assessment Cell Members:

Date Completed: _____

Student Self-Assessment

Complete the following tasks before requesting instructor verification.

Task	Completed
Oracle VirtualBox installed	☐
Kali Linux VM imported	☐
Kali Linux virtual machine boots successfully	☐
Internet connectivity verified	☐
Terminal application operational	☐

Demonstration

Demonstrate the following to your instructor.

Demonstration	Verified
Launch Kali Linux	☐
Open Terminal	☐
Execute <code>ip addr</code>	☐
Execute <code>ping google.com</code>	☐

Instructor Skills Verification

Evaluation	Verified
Workstation prepared correctly	☐
Student demonstrated required skills	☐
Ready to proceed to Part B	☐

Evidence to Retain

Retain the following evidence for inclusion in the **Laboratory Setup and Verification Report**.

- Screenshot of the VirtualBox Manager
- Screenshot of the Kali Linux Desktop
- Output of the `ip addr` command
- Output of the `ping google.com` command

Instructor Signature: _____

Verification Date: _____

Part B – Assemble and Configure the IoT Sensor Node

Overview

The IoT sensor node serves as the primary data acquisition device throughout the laboratory exercises. In this activity, you will assemble the ESP32 (or ESP8266) development board with the DHT temperature and humidity sensor, upload the provided telemetry firmware, and verify that the sensor operates correctly. At this stage, the Raspberry Pi gateway has not yet been deployed; therefore, the sensor node will operate in **Local Sensor Mode** by displaying sensor readings through the Serial Monitor.

Expected Outcome

At the end of this activity, you should have:

- ESP32 (or ESP8266) development board assembled with the DHT sensor.
- Arduino IDE configured for the assigned development board.
- Telemetry firmware successfully uploaded.
- Temperature and humidity readings displayed through the Serial Monitor.
- A fully operational IoT sensor node ready for integration with the Raspberry Pi gateway in Part C.

Step 1. Assemble the IoT Sensor Node

Construct the sensor node according to the wiring diagram provided by your instructor. Connect the following components:

Component	Connection
ESP32 / ESP8266 Development Board	USB connection to workstation
DHT11 / DHT22 Sensor	Data, VCC, and GND
USB Cable	Computer to Development Board

Carefully inspect all wiring connections before connecting the USB cable.

Important

Incorrect wiring may prevent the sensor from functioning properly or may damage the development board.

Step 2. Configure the Arduino IDE

Configure the Arduino IDE for the assigned development board.

1. Launch the Arduino IDE.
2. Select the appropriate development board.
3. Select the correct COM port.
4. Verify that the required libraries are installed:
 - ✓ DHT Sensor Library
 - ✓ PubSubClient

Confirm that the Arduino IDE detects the connected development board.

Step 3. Open the Telemetry Firmware

The instructor will provide the telemetry firmware that will be used throughout the laboratory exercises.

1. Open the provided firmware project in the Arduino IDE.
2. Review the program structure.

3. Verify that the target board and COM port remain correctly configured.
At this stage, no code modifications are required.

Step 4. Upload the Firmware

Upload the telemetry firmware to the development board.

1. Select **Verify** to compile the firmware.
2. Resolve any compilation errors before proceeding.
3. Select **Upload**.
4. Wait until the upload process completes successfully.

The Arduino IDE should display a message indicating that the upload has completed.

Step 5. Verify Local Sensor Operation

1. Open the Serial Monitor.
2. Configure the Serial Monitor to **115200 baud**.
3. Press the **RESET** button on the development board if necessary.
4. Observe the startup messages.
5. If the Raspberry Pi gateway is not yet available, the firmware should automatically enter **Local Sensor Mode**.

Example output:

```
Temperature: 28.4 °C    Humidity: 63 %  
Temperature: 28.5 °C    Humidity: 62 %  
Temperature: 28.6 °C    Humidity: 62 %
```

Verify that the displayed temperature and humidity values update periodically.

Step 6. Validate the IoT Sensor Node

Complete the following validation checklist.

Validation Item	Status
Development board detected by Arduino IDE	☐
Firmware compiled successfully	☐
Firmware uploaded successfully	☐
Serial Monitor operational	☐
Sensor readings displayed	☐
Sensor values update periodically	☐

If all validation items are successfully completed, the IoT sensor node is ready for integration with the Raspberry Pi gateway during **Part C**.

Lab 1 - Part B Completion Record

Assessment Cell Members:

Date Completed: _____

Student Self-Assessment

Complete the following tasks before requesting instructor verification.

Task	Completed
ESP32 / ESP8266 development board assembled with DHT sensor	☐
Arduino IDE configured for the assigned development board	☐
Correct COM port selected	☐
Required libraries installed	☐
Telemetry firmware compiled successfully	☐
Telemetry firmware uploaded successfully	☐
Serial Monitor operational at 115200 baud	☐
Temperature readings displayed	☐
Humidity readings displayed	☐
Sensor readings update periodically	☐

Demonstration

Demonstrate the following to your instructor.

Demonstration	Verified
Show the assembled ESP32 / ESP8266 sensor node	☐
Show the Arduino IDE with the assigned board selected	☐
Show the correct COM port	☐
Show the uploaded telemetry firmware running	☐
Open the Serial Monitor at 115200 baud	☐
Demonstrate temperature readings	☐
Demonstrate humidity readings	☐
Demonstrate that readings update periodically	☐

Instructor Skills Verification

Evaluation	Verified
Sensor node assembled correctly	☐
Arduino IDE configured correctly	☐
Firmware uploaded successfully	☐
Student demonstrated local sensor operation	☐
Student demonstrated required skills	☐
Ready to proceed to Part C	☐

Evidence to Retain

Retain the following evidence for inclusion in the **Laboratory Setup and Verification Report**.

- Screenshot of the assembled ESP32 / ESP8266 sensor node
- Screenshot of the Arduino IDE showing the selected board and COM port
- Screenshot of the successful firmware upload
- Screenshot of the Serial Monitor showing temperature and humidity readings

Instructor Signature: _____

Verification Date: _____

Part C – Deploy the IoT Security Testbed

Overview

In this activity, you will deploy the Raspberry Pi IoT gateway and integrate the ESP32 sensor node prepared in Part B.

The instructor will provide a **custom Raspberry Pi OS image** containing the required IoT network, MQTT brokers, and Node-RED telemetry platform. Students will flash the image using Raspberry Pi Imager and operate the Raspberry Pi as a **headless gateway**. The ESP32 used in Part B will also be reflashed with the laboratory telemetry firmware using the **ESP Student Firmware Flasher**. The firmware configuration is assigned according to the assessment cell.

Expected Outcome

At the end of this activity, you should have:

- Raspberry Pi custom image successfully deployed.
- Raspberry Pi gateway operational.
- ESP32 laboratory firmware installed.
- ESP32 connected to the assigned IoT Wi-Fi network.
- MQTT communication verified.
- Node-RED telemetry dashboard accessible.
- Temperature and humidity readings displayed.
- End-to-end IoT communication verified.
- Baseline configuration documented.

Step 1. Flash the Raspberry Pi Image

1. Insert the assigned microSD card into the workstation.
2. Launch **Raspberry Pi Imager**.
3. Select **Choose Device** and select the assigned Raspberry Pi model.
4. Select **Choose OS**.
5. Select **Use custom**.
6. Select the custom Raspberry Pi OS image provided by your instructor.
7. Select **Choose Storage** and select the assigned microSD card.
8. Review the selected image and storage device.
9. Select **Write** and confirm.
10. Wait for the writing and verification process to complete.
11. Safely eject the microSD card.

Important

The custom image already contains the required IoT gateway configuration and services. Do not install or modify additional software.

Step 2. Boot the Raspberry Pi Gateway

1. Insert the prepared microSD card into the Raspberry Pi.
2. Connect the required network and power connections.
3. Power on the Raspberry Pi.
4. Allow the Raspberry Pi to complete its startup process.

The Raspberry Pi operates as a **headless gateway**.

Students do not need to log in to the Raspberry Pi or access it through SSH. The required services start automatically.

Step 3. Reflash the ESP32 Laboratory Firmware

The firmware used in Part B was only intended to verify the basic operation of the ESP32 and DHT sensor.

For the laboratory environment, install the assigned telemetry firmware.

1. Connect the ESP32 to the workstation using USB.
2. Close the Arduino IDE if it is using the ESP32.
3. Open the **ESP Student Firmware Flasher**:
[ESP Student Firmware Flasher](#)
4. Select the appropriate **ESP Model**.
5. Select your assigned **Group**.
6. Select the installed DHT sensor.
7. Select **Connect & Install**.
8. Select the appropriate serial port when prompted.
9. Wait for the firmware installation to complete.

Important

Do not manually modify the Wi-Fi or MQTT configuration. The firmware configuration is determined by the assigned assessment group.

Step 4. Verify Wi-Fi Connectivity

1. Open the ESP32 Serial Monitor.
2. Set the baud rate to **115200**.
3. Reset the ESP32 if necessary.
4. Observe the startup messages.

A successful connection should display information similar to:

```
=== Connecting to WiFi ===
.....
WiFi connected
IP address: 192.168.50.20
Device ID: ESP32-001
```

The actual IP address and device ID depend on the assigned assessment cell.

Verify that the ESP32:

- connects successfully to the IoT Wi-Fi network;
- receives an IP address; and
- displays its device ID.

Step 5. Verify MQTT Communication

After connecting to Wi-Fi, the ESP32 automatically connects to its assigned MQTT broker.

A successful connection should display:

Attempting MQTT connection...connected

The ESP32 should then periodically publish sensor telemetry.

Example:

```
Published: {"device": "ESP32-001", "temp": 28.4, "humidity": 63}
MQTT topic: sensors/ESP32-001/telemetry
```

Verify that MQTT communication is successful and that telemetry is being published.

Step 6. Access the Node-RED Dashboard

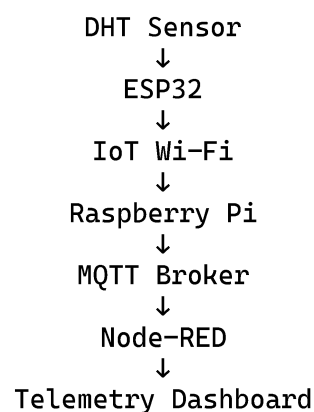
1. Connect the workstation to the laboratory network.
2. Open a web browser.
3. Navigate to the Node-RED dashboard address provided by your instructor.

The dashboard should display telemetry from the assigned ESP32.

Verify that:

- the assigned device is displayed;
- temperature is displayed;
- humidity is displayed; and
- readings update periodically.

The expected telemetry path is:



Step 7. Document the Baseline Configuration

Record the operational configuration of your assessment cell.

Item	Value
Assessment Cell	
ESP32 Device ID	
ESP32 IP Address	
MQTT Broker	
MQTT Port	
MQTT Topic	
Node-RED Address	

This information will serve as the **baseline configuration** for the subsequent security assessment activities.

Final Validation

Before proceeding, verify that:

- Raspberry Pi gateway is operational.
- ESP32 laboratory firmware is installed.
- ESP32 is connected to Wi-Fi.
- MQTT connection is successful.
- MQTT telemetry is being published.
- Node-RED dashboard is accessible.
- Temperature and humidity are displayed.
- Telemetry updates periodically.
- Baseline configuration has been recorded.

Once these conditions are satisfied, the IoT security testbed is ready for the next laboratory activity.

Reflection Questions

1. Why is a standardized Raspberry Pi image important when deploying the IoT security testbed across multiple assessment cells?
2. Why is the Raspberry Pi configured as a headless gateway rather than requiring students to access its operating system directly?
3. What roles do the MQTT broker and Node-RED perform in the IoT testbed?
4. Why is the ESP32 reflashed with the laboratory firmware before it is integrated with the IoT gateway?
5. Why should the baseline configuration be documented before performing security assessment activities?
6. How does verifying the complete telemetry path help establish that the IoT security testbed is ready for security testing?

Assessment Documentation

Lab1 - Setting Up the IoT Security Laboratory Environment

Assessment Cell: _____

Date Performed: _____

Team Members:

A. Assessment Summary

Objective

Set up, configure, and verify the IoT Security Laboratory Environment to ensure that the required assessment workstation, IoT sensor node, Raspberry Pi gateway, MQTT services, and Node-RED telemetry platform are operational before commencing the security assessment activities.

B. Procedures Performed

Briefly describe the activities completed during **Parts A–C** of the laboratory setup.

Part A – Assessment Workstation

- ☐ Installed Oracle VirtualBox
- ☐ Deployed the provided Kali Linux virtual machine
- ☐ Verified Kali Linux operation
- ☐ Verified network connectivity

Part B – IoT Sensor Node

- ☐ Assembled the ESP32/ESP8266 and DHT sensor
- ☐ Configured the Arduino IDE
- ☐ Installed the required libraries
- ☐ Uploaded the sensor functionality firmware
- ☐ Verified temperature readings
- ☐ Verified humidity readings

Part C – IoT Security Testbed

- ☐ Flashed the custom Raspberry Pi OS image
- ☐ Booted and verified the Raspberry Pi gateway
- ☐ Reflashed the ESP32 with the laboratory telemetry firmware
- ☐ Verified ESP32 Wi-Fi connectivity
- ☐ Verified MQTT communication
- ☐ Accessed the Node-RED telemetry dashboard
- ☐ Verified temperature and humidity telemetry
- ☐ Verified end-to-end communication
- ☐ Documented the baseline configuration

Other Activities Performed:

C. Audit Evidence

Attach or reference the evidence collected during Parts A–C.

Evidence	Reference / Filename
Kali Linux Virtual Machine Screenshot	
Kali Linux Network Verification	
ESP32 Sensor Node Photo	
Arduino IDE / Firmware Verification	
ESP32 Local Sensor Readings	
Raspberry Pi Image Installation Evidence	
ESP32 Laboratory Firmware Installation	
ESP32 Wi-Fi Connectivity Evidence	
MQTT Communication Evidence	
Node-RED Telemetry Dashboard Screenshot	
Baseline Configuration Record	
Other	

D. Environment Verification

Record the operational status of the laboratory environment after completing Parts A–C.

Component	Status	Remarks
Kali Linux Workstation	☐ Operational ☐ Issue	
ESP32/DHT Sensor Node	☐ Operational ☐ Issue	
Raspberry Pi Gateway	☐ Operational ☐ Issue	
IoT Wireless Network	☐ Operational ☐ Issue	
MQTT Broker	☐ Operational ☐ Issue	
Node-RED	☐ Operational ☐ Issue	
Telemetry Dashboard	☐ Operational ☐ Issue	
End-to-End IoT Communication	☐ Operational ☐ Issue	

Additional Observations

E. Baseline Configuration

Record the final known-good configuration established during Lab 1.

Configuration Item	Value
Assessment Cell	
Kali Linux Workstation	
Raspberry Pi Gateway	
IoT Network	
ESP32 Device ID	
ESP32 IP Address	
MQTT Broker	
MQTT Port	
MQTT Topic	
Node-RED Address	
Telemetry Dashboard Address	

F. Analysis

Based on the verification results, assess the readiness of the laboratory environment for the succeeding security assessment activities.

Assessment:

G. Recommendations

If issues were identified during the setup or verification, provide the recommended corrective actions before proceeding with the security assessment.

If no issues were identified, indicate:

Laboratory environment ready for assessment.

Recommendations / Corrective Actions:

H. Instructor Validation

Validation Item	Validated
Part A – Kali Linux workstation successfully configured	☐
Part B – ESP32 sensor node successfully assembled and verified	☐
Part C – Raspberry Pi gateway successfully deployed	☐
ESP32 laboratory firmware successfully installed	☐
MQTT communication successfully verified	☐
Node-RED telemetry successfully verified	☐
Required evidence submitted	☐
Baseline configuration documented	☐
Assessment documentation complete	☐
Approved to proceed to Laboratory Exercise 2	☐

Remarks

Instructor Signature: _____

Date: _____